DROWNING IN PLASTIC

INDIA IS NOW THE WORLD'S BIGGEST PLASTIC POLLUTER. WHAT IT MEANS FOR OUR HEALTH AND HOW TO FIX IT

BY SONAL KHETARPAL / *Illustration by* NILANJAN DAS

At the sprawling 250-acre plastic bazaar at Tikri Kalan on the Delhi-Haryana border, towering mounds of white PP raffia bags stuffed with plastic scrap line the dusty, winding lanes. The plastic is segregated into at least 100 distinct varieties, ensuring each sack contains only one kind of plastic, since mixed plastic cannot be processed together. "Plastic waste from every district of India comes here," says Vijay Sharma, office secretary, PVC and Plastic Waste Dealers Association. Between 30 and 50 trucks, each of 10-15 tonne capacity, arrive daily, says Sharma. Other traders estimate the figure to be much higher—200-250 trucks. Which means only one thing: the presence of a massive informal market—much larger than the formal one—right on Tikri Kalan's periphery.

And thereby hangs a whole tale. In a 2022 analysis of Central Pollution Control Board (CPCB) data from 23 states and Union territories, the Centre for Science and Environment (CSE) concluded that only 12 per cent of plastic waste in India is recycled, 20 per cent of it is burned while a significant 70 per cent is unaccounted-for, likely ending up in landfills or dumpyards and along roadsides. No wonder India emerged as the world's largest plastic emitter in a September 2024 study published in science journal *Nature*, which based its conclusions on 2020 data gleaned from 50,702 municipalities worldwide. India contributes 9.3 million tonnes (Mt), or 20 per cent, to the 52.1 Mt of the annual global plastic waste emissions. This refers to the waste emitted into the environment that is not subject to any form of management or control. An estimated 5.8 Mt of plastic is openly burned in India, releasing poisonous gases; 3.5 Mt escapes as loose plastic in the environment that eventually enters the oceans. This is not counting the enormous quantum that gets collected and thrown in mismanaged sites, where it eventually decomposes and leaches deadly toxins into the soil. In an e-mail, the authors of the study told INDIA TODAY that the total plastic waste generated in India is 27.8 Mt, far, far higher than the 3.5 Mt the CPCB estimated for 2020 (see *The India Stack*). As the study says, official figures in India do not include rural areas, open burning of uncollected waste or waste recycled by the informal sector.

9.3

MILLION TONNES PER ANNUM

INDIA'S PLASTIC POLLUTION EMISSIONS, THE HIGHEST IN THE WORLD

Source: Nature, *September 2024*

How did India get here? Plastic first came to India in the form of polystyrene circa 1957; it would take a few decades for other forms to be produced in India. The appeal of the material lies in its very name—derived from the Greek word 'plastikos', meaning that which can be shaped or moulded. A synthetic material made from fossil fuels like petroleum or natural gas, plastic is constructed out of long chains of mol-